

B.Tech. Degree II Semester Regular Examination July 2024

EE/EC/CS/IT 23-200-0205B INTRODUCTION TO CYBER PHYSICAL SYSTEMS (2023 Scheme)

Time: 3 Hours

Maximum Marks: 50

Course Outcomes

On successful completion of the course, the students will be able to:

CO1: Understand the features and components of Cyber Physical Systems.

CO2: Understand the elementary constructs of Arduino Software.

CO3: Develop optimal programs and circuits for interfacing various sensors with Arduino.

CO4: Apply Arduino IDE for developing suitable programs for interfacing sensors and actuators with Arduino and Node MCU.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer *ALL* questions)

(5 × 2 = 10)

	Marks	BL	CO	PO
I. (a) Compare active and passive sensors in IoT using a suitable example.	2	L4	1	1
(b) Describe the general structure of an Arduino program.	2	L2	2	1
(c) Contrast the analogRead() function with digitalRead() in Arduino.	2	L4	2	1
(d) Write a NodeMCU code to blink an LED at 1-second intervals.	2	L6	4	3
(e) How do you set up and use the Arduino IDE for programming Arduino boards?	2	L3	2	5

PART B

(4 × 10 = 40)

II. (a) Explain the features of Cyber Physical System.	6	L2	1	1
(b) What are the key characteristics that define the Internet of Things (IoT)?	4	L2	1	1

OR

III. (a) Define Cyber-Physical Systems (CPS). Explain the general structure of CPS.	6	L2	1	1
(b) List and describe some of the key enabling technologies that support IoT implementations.	4	L1	1	1
IV. (a) Compare and contrast the applications of delay(), millis(), micros() and delayMicroseconds() in Arduino projects.	4	L4	2	1
(b) Explain the various types of operators used in Arduino programming.	6	L2	2	3

OR

V. (a) List and explain the basic data types supported by Arduino.	5	L1	2	1
(b) Explain the functionality of these components on an Arduino board:	5	L2	2	3
(i) Crystal oscillator.				
(ii) Voltage regulator.				
(iii) TX RX LEDs.				
(iv) AREF.				

(P.T.O.)

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VI.	(a) Describe how PWM works on Arduino and its applications.	5	L2	2	1
	(b) How do you integrate an ultrasonic sensor with Arduino for distance measurement?	5	L3	3	3
OR					
VII.	(a) Discuss the process of interfacing humidity and temperature sensors with Arduino.	6	L3	3	3
	(b) What are pull-up and pull-down resistors and how do they affect push button switch interfacing?	4	L2	3	1
VIII.	(a) List the main differences between Arduino Uno and Node MCU development boards.	5	L1	4	1
	(b) Explain how to connect NodeMCU to a WiFi network.	5	L2	4	5
OR					
IX.	(a) Write an Arduino code to control a servo motor.	5	L6	3	3
	(b) How can you interface a gas sensor with NodeMCU?	5	L3	4	5

Bloom's Taxonomy Levels

L1 – 14%, L2 – 43%, L3 – 19%, L4 – 14%, L6 – 10%.
